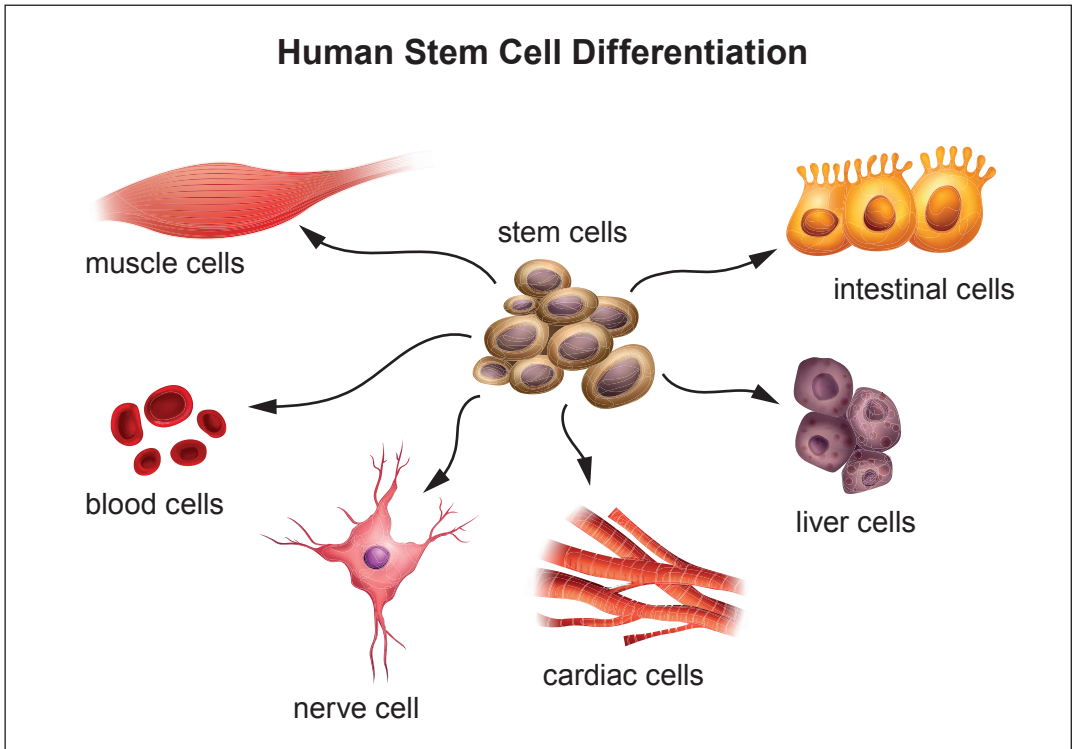


2. The poster is about some uses of stem cells.



(a) Stem cells can differentiate.
Using **one** example from the poster, explain the meaning of differentiate.

[1]

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- (b) Stem cells undergo rapid mitosis.
The table gives five features of cell division. Some features relate to mitosis and some relate to meiosis.
Complete the table by writing **true** or **false** in each row to show the features that relate only to **mitosis**. [3]

Feature of cell division	True or False
daughter cells are genetically identical	
produces four daughter cells	
daughter cells retain the original chromosome number	
daughter cells have chromosomes in pairs	
produces gametes	

- (c) A patient receiving stem cell therapy may be treated with cells taken from their own body or from embryos.
Give **three** advantages of the use of stem cells taken from a patient compared with those from an embryo. [3]

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